Experimental demonstration of reduced fibre solarisation in an oxygen-free atmosphere

The problem

The $^{171}Yb^{+}$ experiments deliver 369.5 nm light through Thorlabs P3-375PM-FC patch cables (PM-S350-HP: PANDA, pure-silica core, FC/APC). After a fresh polish the coupling efficiency is excellent. It then falls over days to weeks, faster at higher power, and both the input and the output end are affected. At 3 mW in the 2.4 µm mode the end faces see $\approx 1.3 \times 10^5$ W/cm^2. Because re-polishing restores the throughput, the damage sits at the end faces (or within micrometres of them), not in the bulk of the fibre. The proposed remedy is to keep the end faces in an **oxygen-free atmosphere**, using **3D-printed cells** with an **oxygen absorber** or an inert-gas purge. The MSc project should demonstrate that this halts the degradation, or find out why it does not.

What the literature says

- **Oxygen cannot reach the bulk.** At room temperature O_2 penetrates silica by only nanometres per year [1], whereas H_2 crosses a fibre in days [2]. Any oxygen effect must act at the end faces, which is where the group's fibres fail.
- **Hydrocarbon contamination: oxygen protects.** Space lasers (Aeolus/ALADIN), UV frequency-conversion cavities in ion-trap labs and GaN laser diodes all suffer absorbing carbon deposits in nitrogen or vacuum, and are run with oxygen for this reason [3–6]. Under CW light at an intensity like ours, deposits grew fastest at *trace* oxygen levels [7].
- **Siloxane contamination: oxygen can harm.** Silicones oxidise to non-volatile SiO_x. At a 410 nm fibre end face, $O_2 \geq 100$ ppm *accelerated* such deposits [8]. This is the strongest support for the group's hypothesis.
- **Oxygen-independent mechanisms.** Particles pulled into the beam, prevented at 313 nm by a dry N_2 overpressure [9]. Laser-induced periodic structures growing out of the silica itself, seen at 405 nm on both faces of small-core fibres [10].
- **The gap.** Nobody has varied O_2 in a controlled way for near-UV fibre end faces, and there are no lifetime data for step-index pure-silica PM fibre at 369.5 nm.

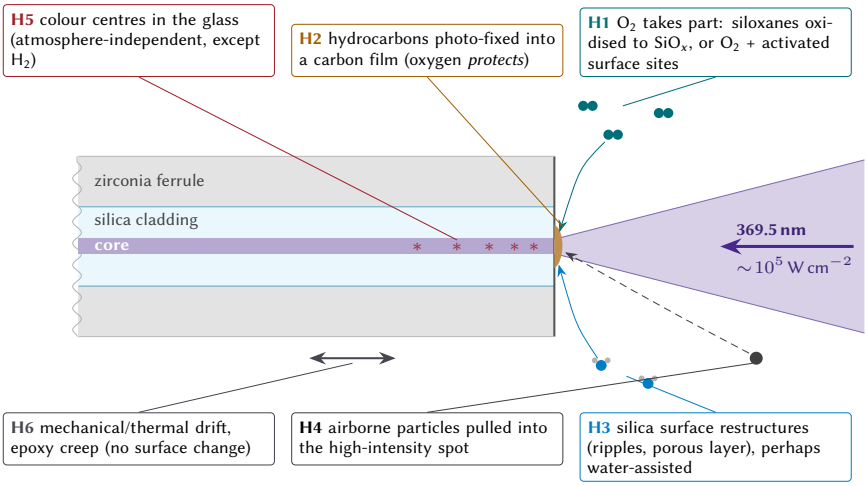
Aim and research questions

Determine whether, and in which atmosphere, the degradation can be stopped, and demonstrate a low-cost 3D-printed enclosure that does it in routine use. **RQ1** Where is the loss, and what is it? **RQ2** Does removing O_2 change the rate, in which direction, and is there a threshold? **RQ3** What are the roles of water, organic and siloxane vapours, and particles? **RQ4** Does a sealed absorber cell hold its atmosphere for months and protect a working link?

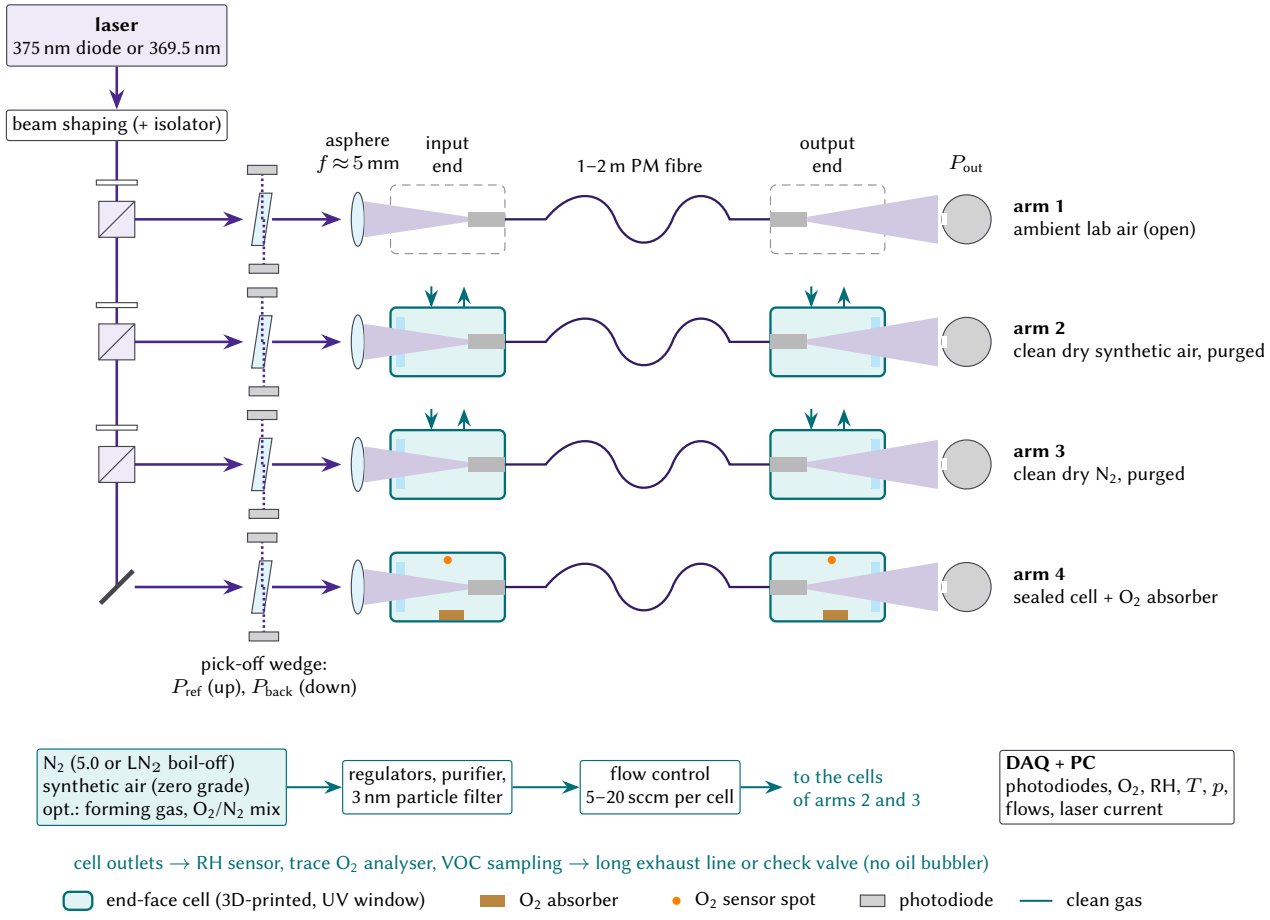
Hypotheses and the decisive test

Hypothesis	AIR	CDA	N2	ABS	tell-tale sign
H1 O_2-assisted (e.g. siloxane → SiO_x)	●	●	○	○	rate rises with O_2; Si/O-rich deposit
H2 carbonaceous contamination, O_2 protects	●	○	●/○	●/↑	C-rich, fluorescent film
H3 restructuring of the silica surface	●	●/○	●/○	●/○	ripples along polarisation
H4 particles	●	○	○	○	discrete particles
H5 colour centres in the glass	●	●	●	●	no surface change
H6 mechanical drift	●	●	●	●	recovered by re-peaking

AIR lab air; **CDA** clean dry synthetic air; **N2** clean dry nitrogen; **ABS** sealed printed cell with O_2 absorber. ● degrades like air, ○ strongly reduced, ↑ faster. **CDA** and **N2** differ *only* in O_2, so comparing them decides the oxygen question.

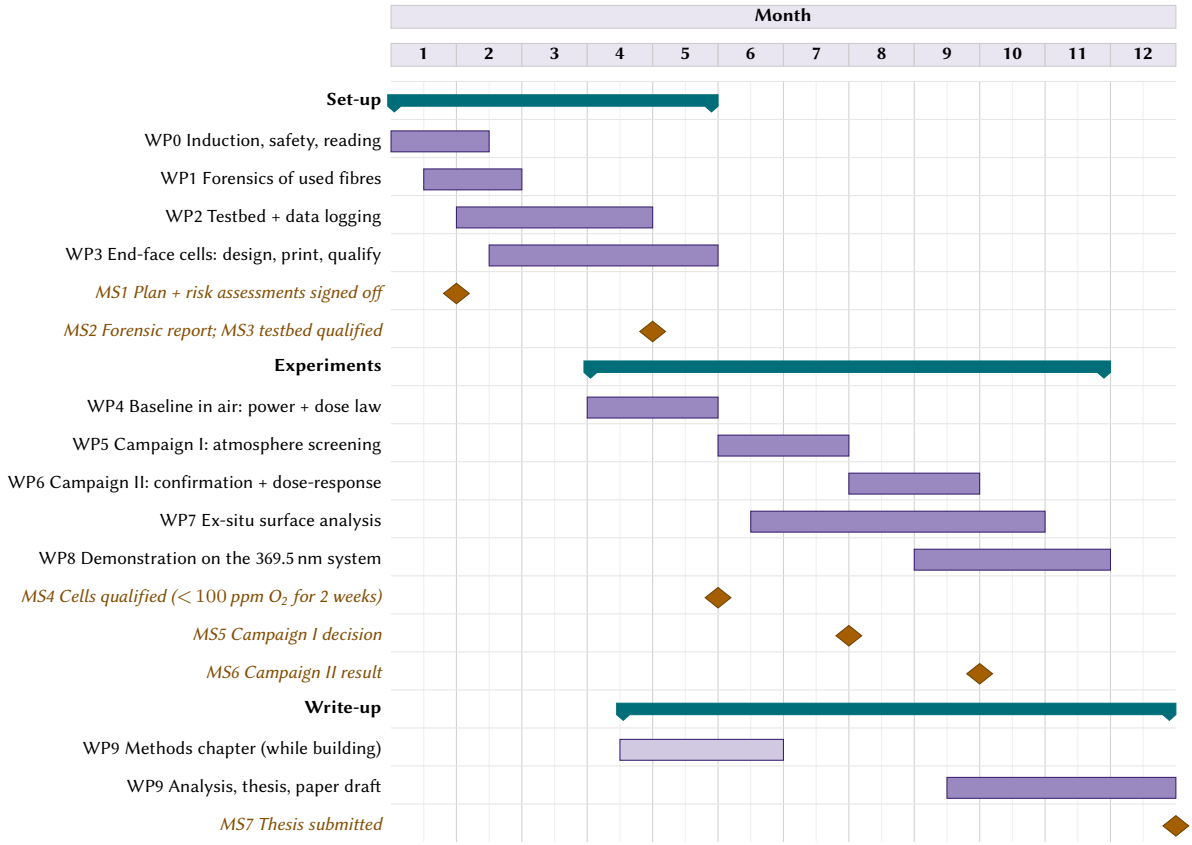


Approach



- **Phase 0, forensics (months 1–2).** Image, test and analyse fibres already degraded in the lab (microscopy, SEM/EDX, AFM, Raman), and survey contaminant sources such as silicones, printed parts and cable jackets.
- **Testbed.** A dedicated 375 nm single-mode diode (70 mW) feeds 4–6 identical arms in *parallel*. For each arm the testbed logs throughput, end-face reflection, O_2, RH, VOC index, temperature and flow continuously. A sentinel arm tracks system drift, and a daily re-peak separates alignment drift from end-face change.
- **End-face cells.** 3D-printed (sealed and baked) with a UV window, the connector fully enclosed, a potted cable gland, gas swept across the face, and an absorber behind a PTFE membrane. An optical O_2 sensor spot reads the cell from outside. Target: < 100 ppm O_2 for months, leak rate $< 1 \times 10^{-5}$ mbar L/s.
- **Campaigns.** Baseline in air at 1, 3 and 10 mW (power law). Campaign I: **AIR**, **CDA**, **N2**, **ABS**. Campaign II: replicates, O_2 dose–response (10 ppm to 1 %), humid gas, polymer and silicone challenges, crossed input/output. Finally, demonstration on a working 369.5 nm link for ≥ 8 weeks.
- **Locating the loss.** At the end of each run, a fixed cleaning sequence (dry, solvent, O_2 plasma or UV/ozone, re-polish) is applied to each face in turn, with a measurement after every step. Uncleaned twins go to surface analysis.

Work plan (12 months; a 6-month variant exists)



Success criteria and deliverables

- The effect of O_2 is measured with a confidence interval: a ratio of rates between **N2** and **CDA** different from 1 by a factor of at least 2 in two runs, or excluded.
- "Halted" means a 95 % upper limit below 1 % per 1000 h at 3 mW (less than 10 % per year of continuous use), at least 20 times below the air arm, with unchanged end faces. This is reachable in 3–6 weeks at 0.3 % daily scatter.
- A qualified cell design (CAD, bill of materials, SOPs) tested on a working link; data, code, thesis, and a short paper if the results allow.

Resources, risks and first steps

Resources. The main new items are the 375 nm diode with mount and driver, 8–12 P3-375PM-FC cables, windows, UV-resistant photodiodes and DAQ, gas purification and flow control, and optical O_2 sensing. Much of the optics, the power meters, the N_2 supply, the 3D printers and the polishing kit are likely to be in the group already. Indicative costs are in the full proposal. **Main risks.** The testbed may not reproduce the lab degradation (mitigated by the forensics first and by tuning the power). Alignment drift may mask the effect (re-peak protocol, reflection monitor). The cells or the printed material itself may contaminate the faces (leak tests, material screening, the **N2+P** arm). Polishing scatter may hide small effects (one procedure, replicates). **First month.** Inductions and risk assessments; the five key papers; hands-on cleaning, inspection and coupling; collecting degraded fibres; a project plan and order list presented to the group.

Key references

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